

SIH26166 — 6-Slide PPT Creation Guide

Smart India Hackathon • Space Technology • Software

This guide compresses the supplied 10-slide PPT creation plan into a strong 6-slide SIH-style deck while preserving the official template's six content sections.

6-slide deck	Use this for
1. Title + Problem	Official title page + one-line problem + stakes
2. Idea / Proposed Solution	Solution, innovation, targeted gap
3. Technical Approach	Architecture + pipeline + one key equation
4. Demo + Results	Baseline vs method + terrain visualization + measured metrics
5. Feasibility, Viability + Impact	Scope, risks, mitigation, reusability, benefits
6. Research + References + Close	Sources, data, final claim, repo/contact

SLIDE 1 — TITLE + PROBLEM

- Put **SIH26166**, the problem title, theme, PS category, Team ID and registered team name.
- Problem in one sentence: multi-instrument lunar imagery has different footprints, illumination and intensity, making reliable registration against external reference imagery difficult.
- Add the key stakes: registration supports mosaicking, DEM generation and landing-site selection.
- Visual: a clean 3-instrument footprint comparison (OHRC / TMC-2 / IIRS) and a small lunar/terrain motif.
- Keep the title page uncluttered. Team details should be readable, not dominant.

Judge-facing focus: Problem clarity + why it matters

SLIDE 2 — IDEA / PROPOSED SOLUTION

- Headline: **Shadow-Invariant Learned Descriptor + Classical SIFT**.
- Explain the exact improvement: classical SIFT/ASIFT can fail under lunar shadow and subtle intensity shifts; the learned descriptor supplies shadow-robust candidate matches.
- Show the workflow: source patch → descriptor network → embedding distance → candidate match → RANSAC verification.
- Mention the training concept: triplet-trained Siamese descriptor using physically rendered synthetic sun-angle pairs from real DEM data.
- Do **not** claim to solve all multi-modal registration. State the single failure mode you target.

Judge-facing focus: What is new and exactly what failure it fixes

SLIDE 3 — TECHNICAL APPROACH

- Use the official technology stack: React + Vite + React Three Fiber; FastAPI + ONNX Runtime; PyTorch; Supabase; Upstash Redis; Sentry; Gemini API.
- Pipeline: **ingest** → **geometry** → **photometric correction** → **SIFT + learned descriptor in parallel** → **RANSAC/homography verification** → **evaluate** → **R3F visualization**.
- Include one simple architecture diagram rather than paragraphs.
- Add exactly **one** defensible equation (for example, the contrastive/triplet loss). Explain in one sentence what it buys you.
- Use a consistent visual language and keep labels large enough for judges to read.

Judge-facing focus: Can this actually work?

SLIDE 4 — DEMO + MEASURED RESULTS

- Show two real scenes side-by-side with visibly different shadow patterns.
- **Left:** classical baseline, where matches scatter/fail under shadow. **Right:** proposed method, where matches hold better.
- Under both, show the R3F terrain view with contour lines so the deck demonstrates a working system, not only a scatter plot.
- Add a compact results table: **RMSE, inlier count, inlier ratio** for baseline vs proposed method.
- If using synthetic DEM-pair validation, label it clearly as **synthetic**. Never mix synthetic and real-data numbers.
- Only show numbers that come from actual logged runs and can be defended in Q&A.;

Judge-facing focus: Show, compare, measure

SLIDE 5 — FEASIBILITY, VIABILITY + IMPACT

- Feasibility: homography, triangulation and RANSAC are closed-form; the learned descriptor is the main trained component.
- Risks from the supplied template: PRADAN registration lag; limited OHRC/reference overlap; DEM dependency in the shadow-consistency check.
- Mitigation: register on PRADAN early; confirm coverage for candidate scenes; independently validate geometry using TMC-2 stereo.
- Impact: reusable pipeline across OHRC/TMC-2/IIRS, supporting mosaicking, DEM generation and landing-site selection.
- Credibility point: name what you deliberately leave out. Honest scope is a strength, not a weakness.

Judge-facing focus: Can it be built and reused?

SLIDE 6 — RESEARCH + REFERENCES + CLOSE

- List the key research reference: Makharia et al., ISRO SAC (2025), after verifying the exact title/DOI before submission.
- List data sources from the supplied template: Chandrayaan Map Browse, ISSDC PRADAN, LROC NAC/QuickMap, JAXA SELENE archive.
- List DEM sources: LOLA / SLDEM2015; crater catalog source if used.
- Close with the actual claim: a working, generic, evaluable registration pipeline across Chandrayaan-2 OHRC/TMC-2/IIRS against external reference imagery, with one measured named improvement.
- Add team/repo/contact at the bottom.
- Final design rule: one idea per slide, one consistent typeface pairing and accent scheme; avoid decorative overload.